MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,
EXCLUSIVE-OR GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-0038510/26A(USAF), dated 15 October 1973.

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TABLE I: Delete and substitute new table I as printed on page 2 of this amendment.

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TABLE II: Delete and substitute new table II as printed on page 3 of this amendment.

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FIGURE 4: Delete and substitute new figure 4 as printed on page 4 of this amendment.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-29)

FSC 5962

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TABLE I. Electrical characteristics tests.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2	-	-	V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V	-	0.7	-	V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.7 V, I_{OH} = -100 µA	2.4	-	±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V V_{IL} = 0.7 V, I_{OL} = 2 mA	-	0.3	±.03	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.4 V	-	20	10X or 2 µA	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	100	N/A	µA
6*	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.3 V	-0.12	-0.36	±.02	mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IH} = 5.5 V V_{IL} = 0 V	-3	-15	-	mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IH} = 5.5 V	-	6.68	-	mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IH} = 4.5 V V_{IL} = 0 V	-	4.4	-	mA

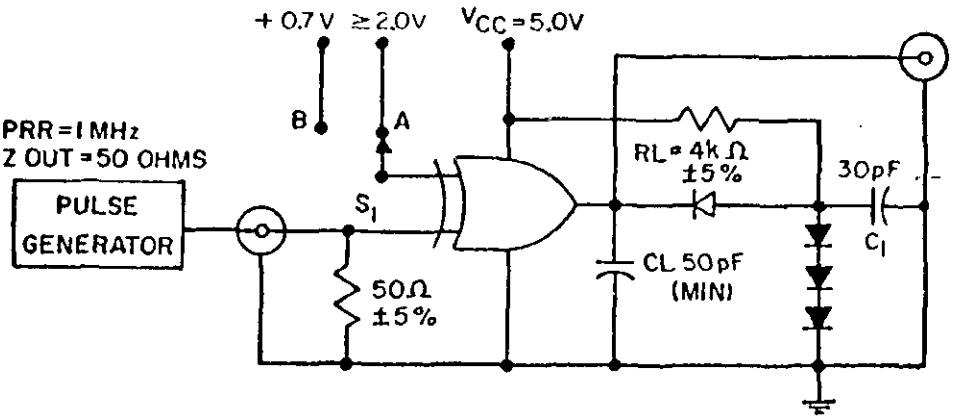
1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

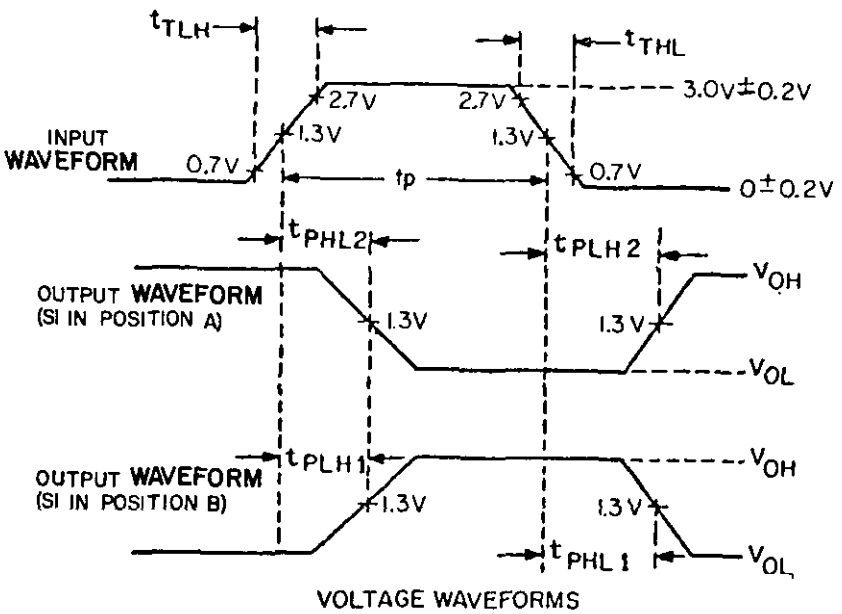
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TABLE II. Switching characteristics.

Characteristic	Symbol	Conditions	Temp.	Limits Min	Limits Max	Units
Propagation delay time to low level output (other input low)	t_{PHL_1}	C_L = 50 pF, R_L = 4 kΩ ±5%, V_{CC} = 5 V, Fig. 4	25°C	7	165	ns
			-55°C, 125°C	7	275	ns
Propagation delay time to high level output (other input low)	t_{PLH_1}		25°C	7	150	ns
			-55°C, 125°C	7	250	ns
Propagation delay time to low level output (other input high)	t_{PHL_2}		25°C	7	66	ns
			-55°C, 125°C	7	110	ns
Propagation delay time to high level output (other input high)	t_{PLH_2}		25°C	7	90	ns
			-55°C, 125°C	7	150	ns

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VOLTAGE WAVEFORMS

NOTES:
1. The generator has the following characteristics: V_{gen} = 3 V, $t_{TLH} = t_{THL} \leq 25$ ns, t_p = 1.0 µs.
2. All diodes are 1N3064, or equivalent.
3. C_L includes probe and jig capacitance.
4. Each gate tested separately.
5. Propagation delay = $\frac{t_{PHL} + t_{PLH}}{2}$

FIGURE 4. Switching test circuit and waveforms.

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